

Petra Bajac



AI Engineer specialized in production LLM systems, with a background in **computational linguistics and NLP**. Builds LLM pipelines, AI agents and data pipelines end to end, from use-case discovery and evaluation to deployment and handover. Previously developed voice assistant prototypes at Mercedes-Benz. Founder of an indie app studio shipping mobile and web products.

Work Experience

AI ENGINEER

WAVESIX LABS | AUGUST 2025-PRESENT | REMOTE

- Designed and delivered **LLM-powered automations, data pipelines and AI assistants** for 10+ clients in various sectors, from use-case discovery to handover.
- **Built data and lead generation pipelines** in Python with LLM-based enrichment, classification and scoring, drawing on sources such as business registers, Apollo and Google Maps.
- Ensured production-grade reliability through automated testing, retries, checkpointing, cost controls and full logging of LLM and API calls.
- **Built AI agents and chatbots** for marketing, content generation, CRM outreach and internal operations.
- **Automated business workflows**, from e-commerce and warehouse data processing to LLM-assisted document review in Microsoft Word.
- **Wrote AI governance documentation** aligned with the EU AI Act and ISO/IEC 42001, and ran AI discovery workshops and use-case prioritization.

FOUNDER & FULL-STACK DEVELOPER

SKARLET FORGE | JANUARY 2026-PRESENT | PART-TIME

- Developed and **released a family of mobile apps** on Google Play, including CI/CD pipelines, staged rollouts and in-app-purchase monetization.
- **Built an AI lead-qualification pipeline for a real-estate client**, combining rule-based and LLM scoring, automated messaging workflows, a sales dashboard and voice-bot integration.
- **Built a 3D-to-2D CAD automation prototype for a manufacturing client**, converting STEP models into fully dimensioned DIN-style engineering drawings using a rule engine with a learn-from-correction feedback loop.

Contact

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[LinkedIn](#)

[GitHub](#)

skarletforge.com

[Google Play](#)

Core skills

PROGRAMMING | PYTHON

Data Analysis

- PyTorch, scikit-learn, Matplotlib, Pandas, NumPy

NLP

- Hugging Face Transformers, spaCy, Embeddings & Semantic Search, Vector Databases, NER, Structured Information Extraction

Generative AI

- LLMs, Prompt Engineering, LangChain, Langdock, OpenAI, Anthropic Claude & Gemini APIs, Azure OpenAI, RAG, AI Agents, LLM Evaluation, ElevenLabs, Multimodal AI

AI ENGINEER

FREELANCE | SEPTEMBER 2024-AUGUST 2025 | REMOTE

- **Developed GenAI solutions** for clients, from rapid prototypes and workflow automations to LLM pipelines, RAG systems and custom agents.

DEVELOPMENT ENGINEER

MERCEDES-BENZ | JUNE 2022-JUNE 2024 | SINDELFINGEN

- **Co-developed the in-house speech prototyping toolchain**, enabling sandboxed voice prototype testing with access to in-vehicle IoT functions.
- **Built prototypes for the HeyMercedes! voice assistant** in this toolchain, designing dialog flows, speech output and decision-table-based intent recognition.
- Fine-tuned the underlying ML model for intent recognition and connected the voice and language backend to frontend UIs in ProtoPie.
- Tested prototypes on benches, in real vehicles and in driving simulators. Prototypes were used for POCs, UX studies and driver distraction studies, with contributions to study design and validation.

DATA ANNOTATOR

FIVERR | AUGUST 2020-APRIL 2022 | STUTTGART

- **Annotated, cleaned and analyzed data** for sentiment analysis and text classification.

DATA ANALYST

TAKELAB | JANUARY 2020-JULY 2020 | ZAGREB, CROATIA

- **Analyzed and annotated text corpora** for climate change research, word sense disambiguation and co-reference resolution, deriving insights for model improvement.

Education

MA ENGLISH AND COMPUTATIONAL LINGUISTICS

OCTOBER 2018-MARCH 2022 | UNIVERSITY OF ZAGREB

- **GPA:** 4.6 on a 5.0 scale.
- **Scholarships:** Scholarship for academic excellence awarded by the City of Karlovac (2018/2019) and the University of Zagreb (2019/2020) - Top 10% of the most successful students.
- **Published Scientific Papers:** Kocijan K., Košković L., Bajac P. (2020). Detecting Hate Speech Online: A Case of Croatian. NooJ 2019, Springer CCIS vol. 1153.

Additional tools & technologies

Web & Mobile

- React, Vite, Tailwind, Capacitor, PySide6

Cloud & Data

- Supabase, PostgreSQL, Cloudflare, Data Warehousing, Web Scraping, GCP

Testing & DevOps

- Playwright, Vitest, k6, CI/CD, Sentry, Docker, Unit Testing, OWASP ZAP

AI Governance

- EU AI Act, ISO/IEC 42001, Data Governance

Other

- REST API, Webhooks, Prototyping, Conversational UX, RegEx, n8n

Languages

- Croatian – Mother tongue
- English – C2
- German – C1
- Dutch – B2
- Russian – A2

Personal Interests

- Powerlifting
- MMA & Grappling
- Photography | Drawing